Peer review and response

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

Before you start

Recall Writing results and limitations. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Explain the mechanism and terminology of peer review and response. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of peer review and response.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Writing results and limitations. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Figures that preserve **evidence**. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Inspecting failures and anomalies. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of peer review and response

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Writing results and limitations. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain peer review and response using the records and boundaries of this fictional investigation.

Notice the evidence

> Investigate the supplied Peer review and response case. Separate observations, given facts and assumptions before choosing an explanation.

Peer review challenges **evidence**, method and clarity. Responses should explain changes with artifacts rather than merely defend the original wording.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Peer review and response. Connect each labelled input, deciding step and result.

For each comment record concern, response, change location and new **evidence** if needed. If no new **evidence** exists, narrow the claim instead of promising an unperformed test.

Peer review and response — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Reviewer asks for a fitted-line baseline; author responds that the neural network is advanced, without a comparison.	For each comment record concern, response, change location and new evidence if needed. If no new evidence exists, narrow the claim instead of promising an unperformed test.	The response lacks evidence. Add a fair comparison under the plan or narrow the claim to the existing baseline; record any new analysis as such.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Peer review and response. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Reviewer asks for a fitted-line baseline; author responds that the neural network is advanced, without a comparison. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: The team only changes "photos" to "plants" without revisiting analysis.

Predicted result

Prediction: The substantive independence issue remains unresolved.

Why the result follows

Reason: Renaming records does not change their shared source objects.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Peer review and response to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Review comment	A concern about evidence, method or presentation
Response record	A mapping from comment to change or justified disagreement
Claim narrowing	Reducing a statement to the evidence actually available

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Peer review and response, predict a step, then compare with its explanation.

Reviewer asks for a fitted-line baseline; author responds that the neural network is advanced, without a comparison.

The response lacks **evidence**. Add a fair comparison under the plan or narrow the claim to the existing baseline; record any new analysis as such.

Worked explanation · Peer review and response

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

> The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Peer review and response

Respond to a reviewer with **evidence**. A reviewer asks why four photos of one plant were treated as four independent plants.

The facts we will use

> #### Supplied facts
>
> The original table contains four views of P1 and four views of P2: eight photos from two plants.

AI ACADEMY / AI-7/32

Peer review and response

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify the concern

The reviewer questions the unit of analysis.



2. Check the records

Plant identifiers confirm two plants, not eight independent plants.



3. Revise transparently

Correct the description and revise any analysis that assumed independence.

CHANGE ONE THING

The team only changes "photos" to "plants" without revisiting analysis.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Identify the concern	The reviewer questions the unit of analysis.
2. Check the records	Plant identifiers confirm two plants, not eight independent plants.
3. Revise transparently	Correct the description and revise any analysis that assumed independence.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Peer review should connect a criticism to the actual records and a concrete correction or reasoned response. Polite language alone cannot repair a methodological error.

What this example does not establish

Keep the stated limit

Two plants provide limited coverage; grouping correctly does not create more independent **evidence**.

Keep the stated limit

Read the labelled picture: Count photos and underlying plants in the reviewer’s concern. What analysis assumption needs revision, and why does changing only the word photos to plants fail to repair it?

Keep the stated limit

Change one thing: The team only changes "photos" to "plants" without revisiting analysis. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Peer review and response claim and explain a repair using the supplied evidence.

Agreement with a reviewer is not required, but a disagreement needs explicit reasoning and **evidence**.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: Agreement with a reviewer is not required, but a disagreement needs explicit reasoning and **evidence**.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Peer review and response with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Peer review and response”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Peer review and response. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Count photos and underlying plants in the reviewer's concern. What analysis assumption needs revision, and why does changing only the word photos to plants fail to repair it?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Peer review and response. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Reviewer asks for a fitted-line baseline; author responds that the neural network is advanced, without a comparison. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or "not run"	

Record	Your evidence
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Peer review and response. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: Agreement with a reviewer is not required, but a disagreement needs explicit reasoning and **evidence**.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Reproducible AI Research Study. Keep an annotated decision and the evidence supporting it.

Add a labelled peer review and response **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of peer review and response.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Peer review and response case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Peer review and response — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Reviewer asks for a fitted-line baseline; author responds that the neural network is advanced, without a comparison.	For each comment record concern, response, change location and new evidence if needed. If no new evidence exists, narrow the claim instead of promising an unperformed test.	The response lacks evidence. Add a fair comparison under the plan or narrow the claim to the existing baseline; record any new analysis as such.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Reviewer notes missing failure counts; author deletes the limitation paragraph. What is an adequate response?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

> Reviewer notes missing failure counts; author deletes the limitation paragraph. What is an adequate response?

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Peer review and response, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Respond to a reviewer with **evidence**. A reviewer asks why four photos of one plant were treated as four independent plants.

By the end, I can	How I will show it
Explain the mechanism and terminology of peer review and response	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Peer review and response without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of peer review and response

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Respond to a reviewer with **evidence**. A reviewer asks why four photos of one plant were treated as four independent plants.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Correct the description and revise any analysis that assumed independence.

1. Revise transparently

2. Check the records

3. Identify the concern

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Revise transparently Correct the description and revise any analysis that assumed independence.

Peer review should connect a criticism to the actual records and a concrete correction or reasoned response. Polite language alone cannot repair a methodological error.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The team only changes "photos" to "plants" without revisiting analysis. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. The substantive independence issue remains unresolved.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The substantive independence issue remains unresolved. Renaming records does not change their shared source objects.

Repair a missing building block

Review Writing results and limitations (ai-7/31). Goal: Explain the **mechanism** and terminology of writing results and limitations. Short practice: Compute the supported-answer rates for 9/12 and 10/12.

What actual outcome was measured, and why is a claim that every student becomes smarter outside this **evidence**?. Return to this week and explain what you changed.

Review Figures that preserve **evidence** (ai-7/29). Goal: Explain the **mechanism** and terminology of figures that preserve **evidence**. Short practice: Draw count bars for 6/10 and 8/10 using the stated axis.

Why must the denominator and fixed task context remain visible, and what distortion arises from an undisclosed axis start at five?. Return to this week and explain what you changed.

Review Inspecting failures and anomalies (ai-7/26). Goal: Explain the **mechanism** and terminology of inspecting failures and anomalies. Short practice: Calculate mean and median for 2,2,2,10.

What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

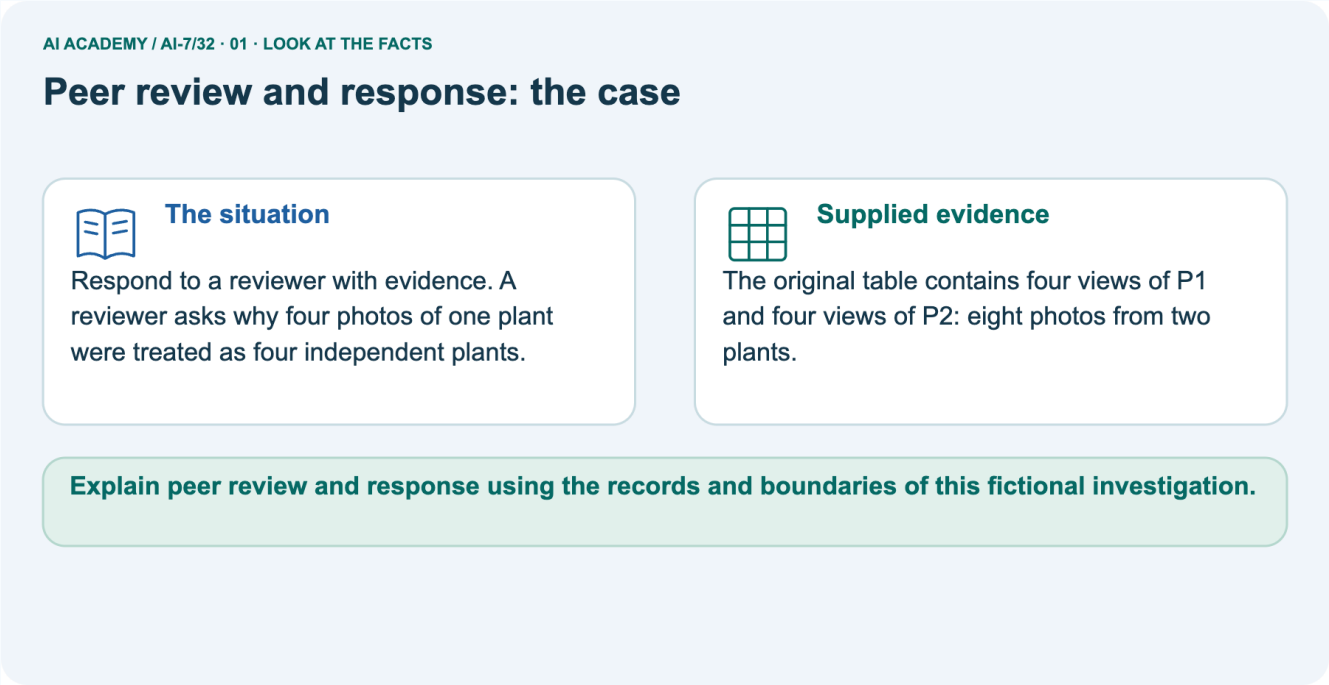
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Peer review and response graphic using its labels, deciding evidence and relevant limitation.

01. Peer review and response: the case



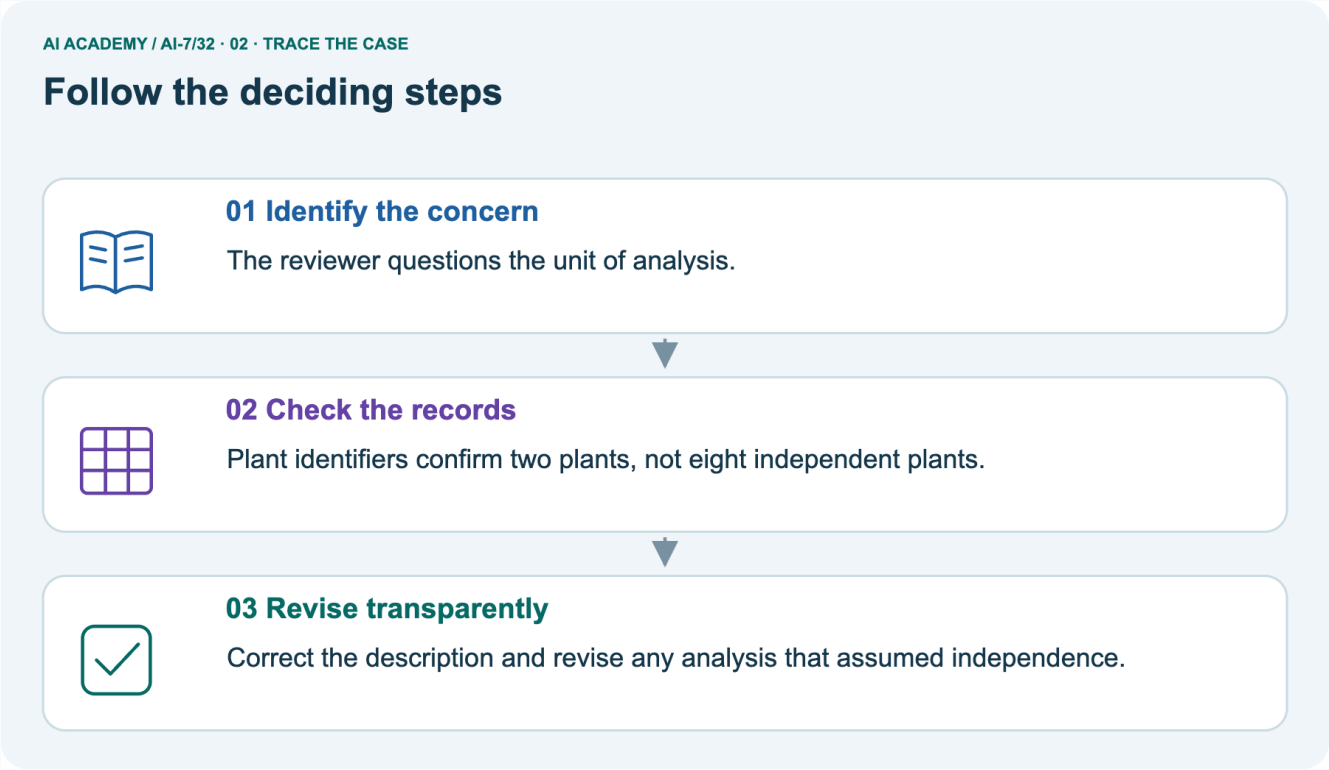
Follow the labelled picture with the matching explanation. Open the vector diagram

Respond to a reviewer with **evidence**. A reviewer asks why four photos of one plant were treated as four independent plants.

Supplied facts: The original table contains four views of P1 and four views of P2: eight photos from two plants.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

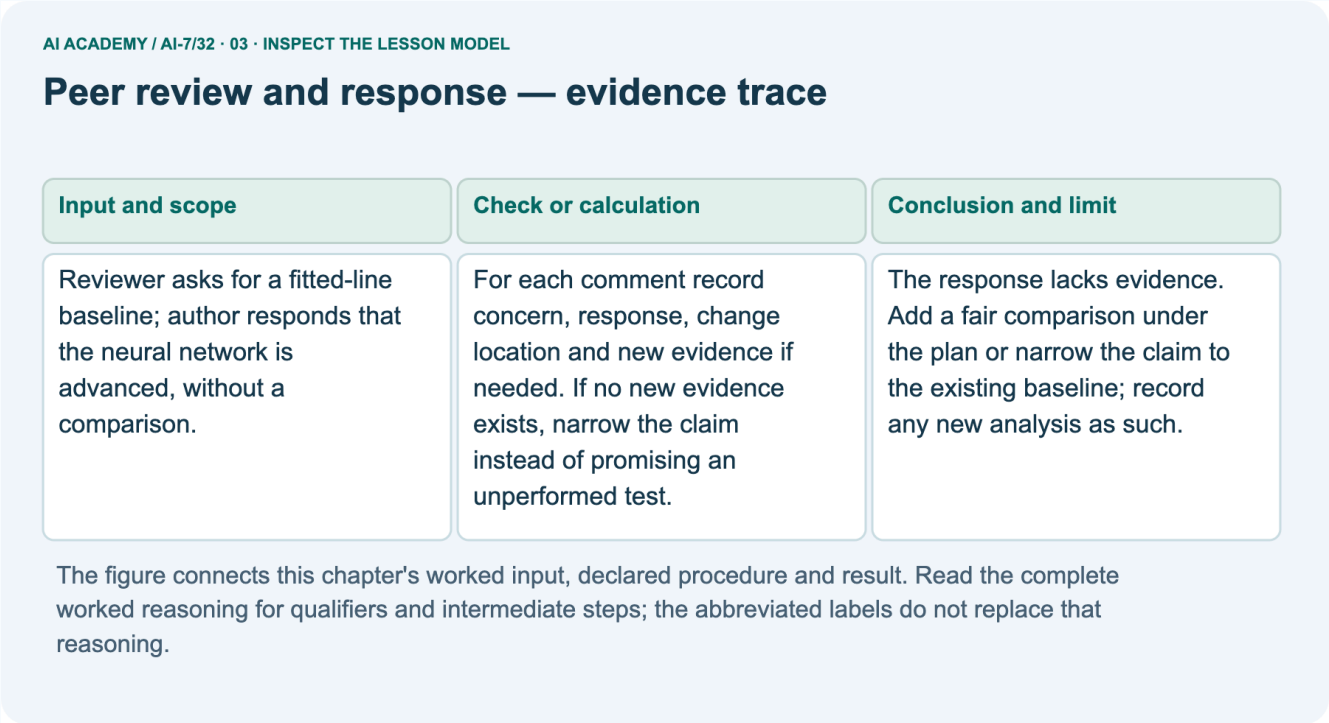


Follow the labelled picture with the matching explanation. Open the vector diagram

Identify the concern: The reviewer questions the unit of analysis. → Check the records: Plant identifiers confirm two plants, not eight independent plants. → Revise transparently: Correct the description and revise any analysis that assumed independence.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Peer review and response — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Peer review and response — **evidence** trace.

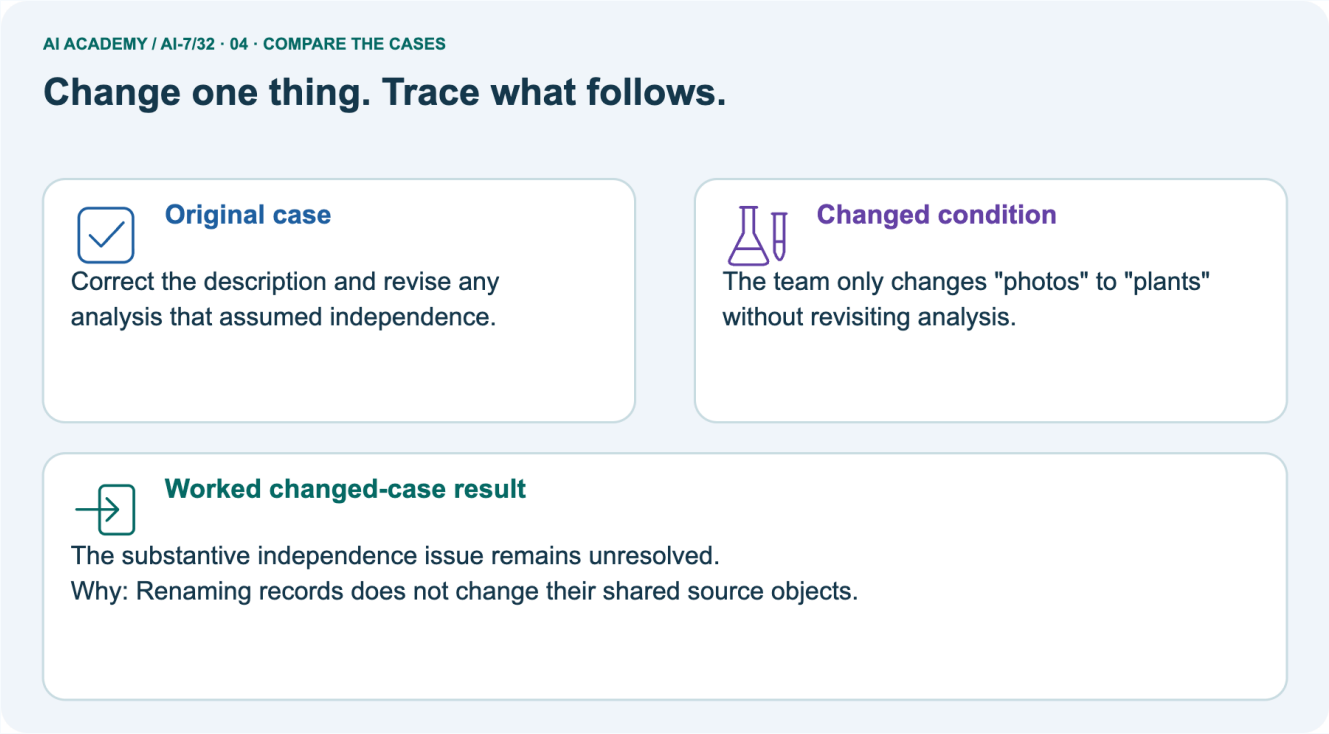
Reviewer asks for a fitted-line baseline; author responds that the neural network is advanced, without a comparison. | For each comment record concern, response, change location and new **evidence** if needed.

If no new **evidence** exists, narrow the claim instead of promising an unperformed test. | The response lacks **evidence**.

Add a fair comparison under the plan or narrow the claim to the existing baseline; record any new analysis as such.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



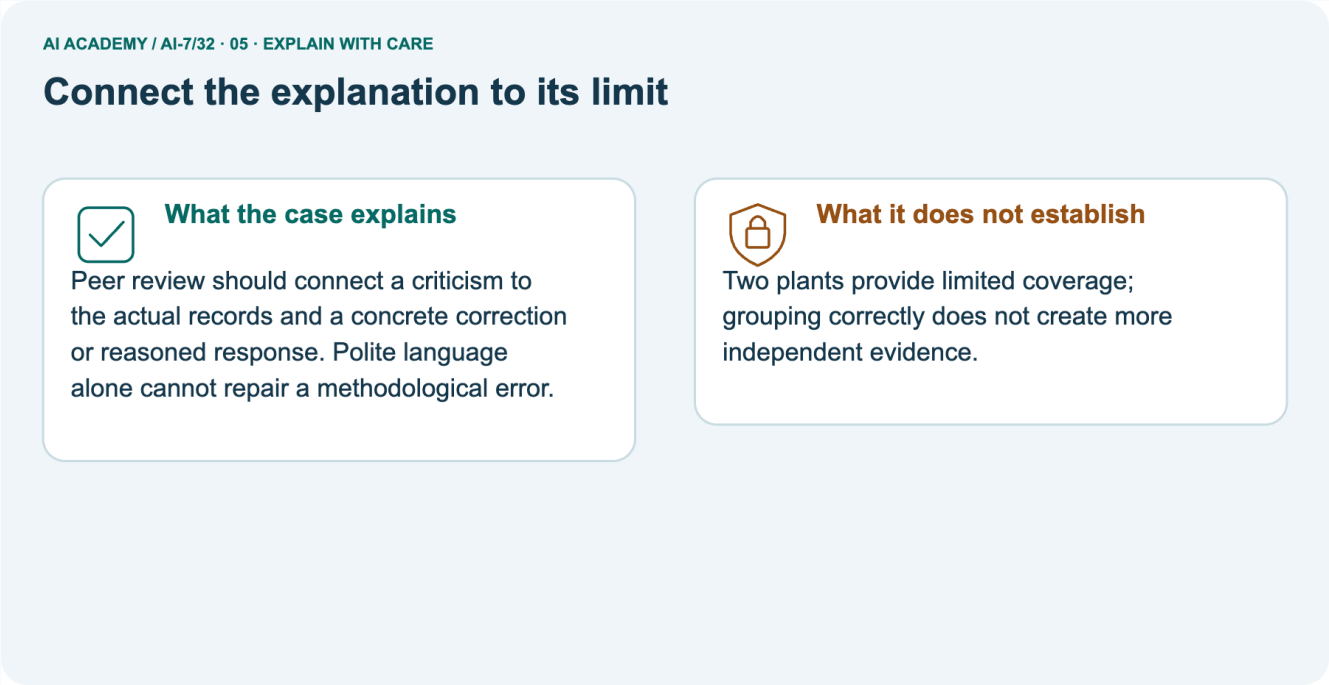
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Correct the description and revise any analysis that assumed independence.
Changed condition: The team only changes "photos" to "plants" without revisiting analysis.

Expected result: The substantive independence issue remains unresolved. Reason: Renaming records does not change their shared source objects.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



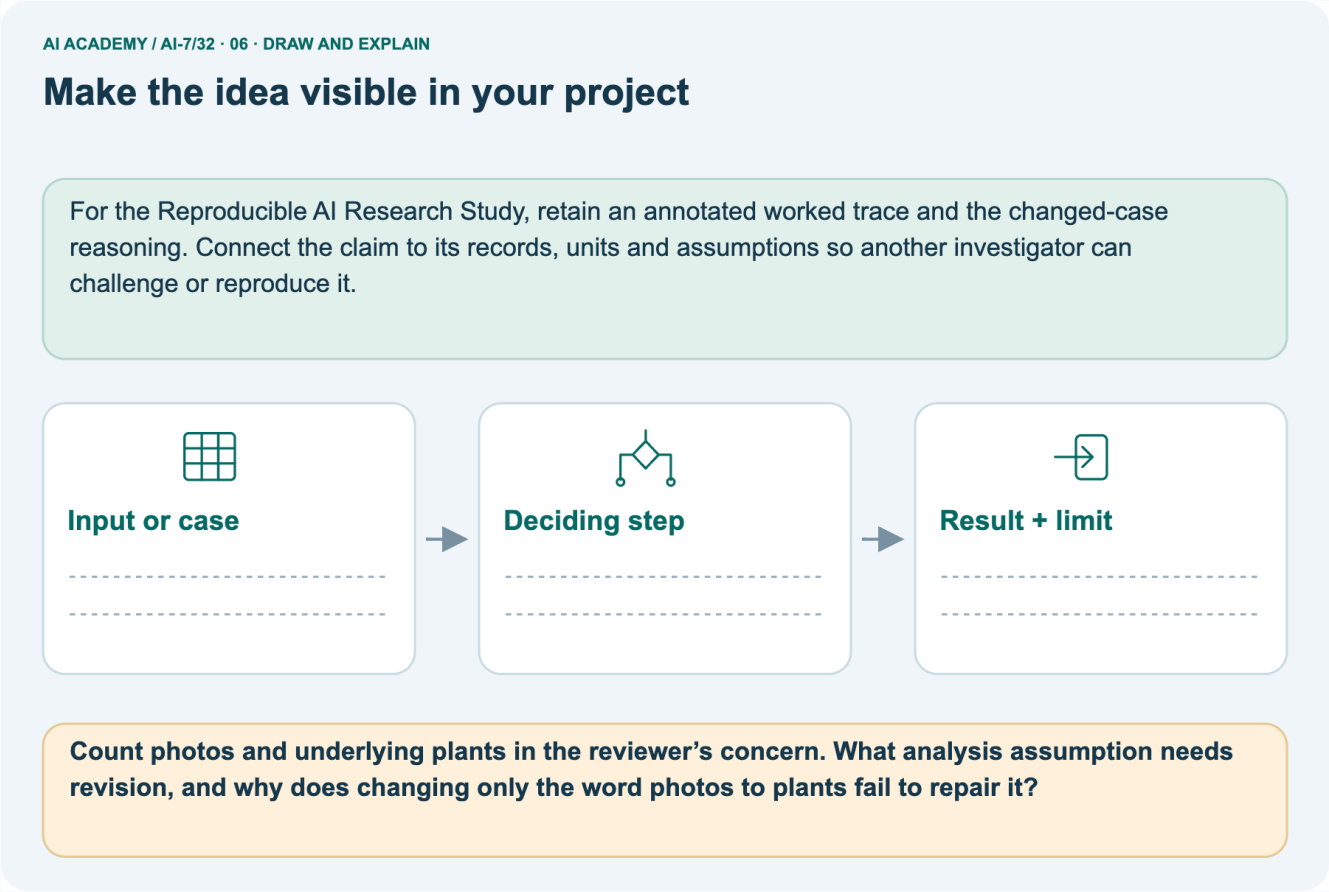
Follow the labelled picture with the matching explanation. Open the vector diagram

Peer review should connect a criticism to the actual records and a concrete correction or reasoned response. Polite language alone cannot repair a methodological error.

Limit: Two plants provide limited coverage; grouping correctly does not create more independent **evidence**.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Count photos and underlying plants in the reviewer's concern.

What analysis assumption needs revision, and why does changing only the word photos to plants fail to repair it?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.